

CSE260 Summer 2025

Assignment 3

Total: 20

This assignment must be handwritten. Show ALL steps in ALL questions.

1. [4.5 Marks] Build a calculator that can add or subtract two 4-bit numbers, X and Y. If both numbers are **greater than or equal to 12**, the calculator will perform **X+Y** (addition). Otherwise, it will perform **X-Y** (subtraction). Design the circuit using a **4-bit parallel adder-subtractor**. Use external gates if required.
2. [4.5 Marks] Build a calculator that can add or subtract two 4-bit numbers, A and B. The calculator will perform **A+B** if B is **divisible by 12**; otherwise, it will perform **A-B**. Design the circuit using a **4-bit parallel adder-subtractor**. Use external gates if required.
3. [3 Marks] Design a circuit that takes two **3-bit** numbers, A and B, and performs **2A - B** using a **4-bit parallel adder**, incorporating only additional **XOR** gates as necessary.
4. [4 Marks] Implement the following Boolean function using a single **2:1 MUX**:
 $F(A,B,C) = \sum(1,3,4,7)$. Use external gates if required.
5. [4 Marks] Implement the following Boolean function using only **2:4 decoder(s)**:
 $F(A,B,C,D) = \sum(0,1,2,7,10,11,14)$. You may use only external **NOR** gates if necessary.